

Soumendu Majee

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RESEARCH INTERESTS **Computational Imaging, Computed Tomography, and Machine Learning**

EDUCATION **Ph.D.**, Electrical and Computer Engineering June 2021
Purdue University, IN, USA
Advisors: Prof. Charles A. Bouman and Prof. Gregory T. Buzzard
Thesis: High Speed Imaging via Advanced Modeling
GPA: 4.0/4.0

B.Tech., Electronics and Electrical Communication Engineering May 2014
Indian Institute of Technology, Kharagpur, India
GPA: 9.1/10

SKILLS

- **Programming languages:** Python, Cython, MATLAB, C, C++, C#, Bash, L^AT_EX
- **Libraries/Packages/Tools:** Pytorch, Tensorflow, Keras, OpenCV, scikit-learn, OpenMP, MPI, Git, ADB

EXPERIENCE **Staff Research Engineer** March 2024 - Present
Senior Engineer, Research April 2022 - February 2024
MPI Lab, Samsung Research America, TX, USA

- Led the university research collaborations with Rice University
- Led the development and commercialization of Unified-Tone-Map across different camera pipelines for Samsung Galaxy S25
- Worked on the development and commercialization of Tele Tetra AI Multiframe Zoom for Samsung Galaxy S24
- Worked on the commercialization of Tetra High-resolution pipeline for Samsung Galaxy S24
- Led the development of Under Display Camera pipeline for Samsung Galaxy Z Fold5
- Worked on the development and commercialization of Under Display Camera for Samsung Galaxy Z Fold4

Postdoctoral Research Associate July 2021 - April 2022
Los Alamos National Laboratory, NM, USA

- Material Identification from Radiographs and Intrinsic Radiation
- Learning Noise-robust Features for Dynamic Reconstruction

Research Aide July 2019 - Sept 2019
Advanced Photon Source, Argonne National Laboratory, IL, USA

- Coded Exposure for High Speed X-ray CT Imaging
 - Developed CodEx: a synergistic combination of coded acquisition and reconstruction for high speed tomographic imaging

Graduate Research Assistant Jan 2015 - June 2021
Purdue University, IN, USA

- Multi-Slice Fusion for 4D X-ray CT Reconstruction
 - Developed a novel method “Multi-Slice Fusion” that fuses multiple low-D Convolutional Neural Networks to implement a 4D image prior

- Denoising Short-exposure Dynamic Radiographs
 - Developed a deep-learning based transfer-learning method for denoising short-exposure dynamic radiographs
- Multi-Domain Weighing for Metal Artifact Reduction in X-ray CT
 - Developed a novel data and image domain weighing for reducing metal artifacts in X-ray CT
- Multi-Orientation Fusion CT reconstruction
 - Developed a modular method to fuse information from multiple CT scans at different orientations to produce a joint reconstruction
- Cone-beam X-ray CT reconstruction of Additively Manufactured Parts
 - Developed a high fidelity CT reconstruction method using scatter correction for defect detection in additively manufactured Parts
- Detection and Localization of Neurons in Fluorescence Microscopy Images
 - Developed novel neuron detection method to improve frame-rate of fluorescence microscopy neuron imaging by selectively scanning neuron locations.

Undergraduate Researcher
Indian Institute of Technology, Kharagpur, India

Jan 2014 - August 2014

- Efficient Wideband Spectrum Sensing for Cognitive Radio
 - Developed a novel spectrum estimation algorithm for spectrum sensing in cognitive radio

Research Intern
University of Toronto, ON, Canada

May 2013 - July 2013

- Detection of Human Thought from EEG (Electroencephalograph) Signals

SELECTED PUBLICATIONS Zhang, M., Majee, S., Wang, C., Lee, S.J. and Sheikh, H., 2024. CoDISP: Exploring Compressed Domain Camera ISP with RGB-guided Encoder. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 5878-5888).

Yang, D., Kemp, C.A., Majee, S., Buzzard, G.T. and Bouman, C.A., 2024. Pixel-weighted Multi-pose Fusion for Metal Artifact Reduction in X-ray Computed Tomography. arXiv preprint arXiv:2406.17897.

Balke, T., Davis Rivera, F., Garcia-Cardona, C., Majee, S., McCann, M.T., Pfister, L. and Wohlberg, B.E., 2022. Scientific Computational Imaging COde (SCICO). Journal of Open Source Software, 7(LA-UR-22-28555).

Majee, S., Aslan, S., Gursoy, D. and Bouman, C.A., 2022. CodEx: a modular framework for joint temporal de-blurring and tomographic reconstruction. IEEE Transactions on Computational Imaging, 8, pp.666-678.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2021. Multi-slice fusion for sparse-view and limited-angle 4D CT reconstruction. IEEE Transactions on Computational Imaging, 7, pp.448-462.

Balke, T., Majee, S., Buzzard, G.T., Poveromo, S., Howard, P., Groeber, M.A., McClure, J. and Bouman, C.A., 2018. Model-Based Cone-Beam Tomography with Scatter Correction. Manuscript in preparation for IEEE Transactions on Computational Imaging.

Majee, S., Balke, T., Kemp, C.A., Buzzard, G.T. and Bouman, C.A., 2019, May. 4D X-ray CT reconstruction using multi-slice fusion. In 2019 IEEE International Conference on Computational Photography (ICCP) (pp. 1-8). IEEE.

Balke, T., Majee, S., Buzzard, G.T., Poveromo, S., Howard, P., Groeber, M.A., McClure, J. and Bouman, C.A., 2018. Separable models for cone-beam MBIR reconstruction. electronic imaging, 2018(15), pp.181-1.

Majee, S., Ye, D.H., Buzzard, G.T. and Bouman, C.A., 2017. A model based neuron detection approach using sparse location priors. Electronic Imaging, 2017(17), pp.10-17.

Majee, S., Ray, P. and Cheng, Q., 2015, November. Efficient wideband spectrum sensing using random projection. In 2015 49th Asilomar Conference on Signals, Systems and Computers (pp. 141-145). IEEE.

OPEN SOURCE SOFTWARE CONTRIBUTIONS Super-Voxel Model Based Iterative Reconstruction (SVMBIR), Software library available from <https://github.com/cabouman/svmbir>. Docs: <https://svmbir.readthedocs.io/en/latest/overview.html>
Model-Based Iterative Reconstruction for cone-beam (MBIRCONe), Software library available from <https://github.com/cabouman/mbircone>. Docs: <https://mbircone.readthedocs.io/en/latest/overview.html#>
"Scientific Computational Imaging COde (SCICO)", Software library available from <https://github.com/lanl/scico>. Docs: <https://scico.readthedocs.io/en/stable/overview.html>

INVITED TALKS	Electronic Imaging	January 2023
	Samsung Research America	January 2022
	Oak Ridge National Laboratory	March 2021
	Los Alamos National Laboratory	February 2021
	Department Of Energy Light-source Tomography Coordination Meeting	February 2021
	Electronic Imaging	January 2021
	Electronic Imaging	January 2019

PROFESSIONAL Paper Reviewer

VOLUNTEERING IEEE Transactions on Image Processing
IEEE Transactions on Computational Imaging
IEEE International Conference on Image Processing
International Journal of Medical Physics Research and Practice
Journal of Nondestructive Evaluation

TEACHING EXPERIENCE	Teaching Assistant Purdue University Linear Circuits Analysis (ECE 201) Digital Image Processing I (ECE 637)	Fall 2014, Spring 2016, Spring 2017
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SCHOLARSHIPS & AWARDS	- Received president's award for contributions in High-Mega-Pixel AI Multi-Frame Processing - Received CTO award for contributions in AI Multi-frame Zoom for Galaxy S24 Ultra - Awarded MITACS Globalink Research Fellowship - Awarded the Jagadis Bose National Science Talent Search Scholarship - Ranked 1 in the WBJEE Engineering Entrance Examination	2024 2024 2013 2010 - 2014 2010
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REFERENCES Available upon request